$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - \int_0^\infty \mathrm{d}\tau \,\Big\{\nu(\tau)\big[\hat{X}, \big[\hat{X}(-\tau), \hat{\rho}_{\mathcal{S}}(t)\big]\big] \\ -\mathrm{i}\eta(\tau)\big[\hat{X}, \big\{\hat{X}(-\tau), \hat{\rho}_{\mathcal{S}}(t)\big\}\big]\Big\}. \quad (5.22)$$

We see that the decohering (and dissipative) influence of the environment is completely described by integrals over the noise and dissipation kernels (5.16) and (5.17) multiplied by products of (Schrödinger-picture and interaction-